Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Versatile Software Engineer and AI practitioner with 8+ years of full-stack production engineering across PropTech, healthcare, and research domains. At Entrata (100K+ MAU), shipped 20+ production modules - financial services automation, compliance systems, data pipelines, cloud migration, and microservices architecture across 120+ Agile sprints. Built agentic AI systems independently: Leasel (8-agent LangGraph platform with retrieval-augmented generation and human-in-the-loop design) and AgentProbe (open-source AI testing toolkit with CI-integrated evaluation engine). Strong foundation in data analytics through M.S. Data Analytics Engineering at George Mason University (GPA 3.87) with hands-on ML, statistical modeling, and research pipeline experience. Equally effective shipping production systems, designing data architectures, building AI-powered applications, or leading cross-functional technical delivery.

CORE COMPETENCIES

Full-Stack Engineering | AI Agent Development | Data Analytics & ML | Cloud Architecture (AWS) | Multi-Agent Orchestration | SQL & Database Design | Compliance & Regulated Data | Agile Delivery | RAG & LLM Systems | Cross-functional Collaboration

WORK EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built an agentic mission-planning system as a LangGraph stateful graph with conditional routing - compliant missions export, non-compliant route to rejection with grounded, cited reasons.
- Implemented a RAG layer over FAA Part 107 regulatory corpus; engineered validation guardrails (altitude limits, battery budgets, no-fly-zone checks) blocking non-compliant outputs before export.
- Improved agent accuracy through iterative prompt engineering and retrieval tuning; parsed natural-language inputs into structured flight plans using photogrammetry math.

George Mason University - College of Engineering

Aug 2025 - May 2026

Graduate Teaching Assistant (Part-time)

- Delivered technical workshops to 600+ students across MIS 303 sections covering Python programming, database design, SQL, and data applications; provided one-on-one debugging and logic support.
- Developed supplementary exercises and rubric-aligned feedback; coordinated content delivery with lead instructor across all sections.

George Mason University - Research

Aug 2024 - Aug 2025

Research Associate

- Built multi-source prediction models (Python, R) achieving $R^2 = 0.977$ across 190+ countries on WHO and UK Road Safety datasets - full pipeline: data collection, transformation, hypothesis testing, and validation.
- Applied Isolation Forest anomaly detection across 3 research pipelines; surfaced anomalies manual review missed, improving downstream model reliability.
- Designed faculty leadership dashboards in Tableau, ggplot2, and Quarto - improved research comprehension 40% across two cohorts.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer (Remote)

- **Shipped 20+ full-stack production modules** (PHP, Python, React, PostgreSQL, RabbitMQ) across 120+ Agile sprints for 100K+ monthly active users - 99.5% uptime over 6+ consecutive months.
- **Managed end-to-end financial services functionalities**: HAP allocation, AMI-based rent pricing, HUD payments, tax credit calculations, ledger management, and income eligibility verification - zero compliance violations across federal and state housing regulations.
- **Shipped first production AI pipeline** across 1.6M financial records (Python, PyTorch, OpenAI API) - automated data processing replacing manual Excel workflows; output validation gates and quality controls at launch.
- **Led microservices architecture redesign** using Docker - 50% improvement in scalability, 30% reduction in deployment time; converted 20% of product modules into service-based architecture.
- **Achieved 100% PHI compliance** on AWS S3 cloud migration with Libsodium-encrypted pipelines; zero security incidents; updated 50%+ of codebase for encryption compliance.
- **Optimized 200+ SQL queries** across all client environments (30% performance gain) through execution plan tuning, indexing, partitioning, caching, and audit tables.
- Eliminated 80% of manual operational workload via Python/Bash automation; mentored 5+ junior engineers; conducted 300+ peer code reviews.
- Built Salesforce integrations, CRM data pipelines, and workflow automation using Clay and n8n.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer / Trainee

- Developed full-stack property management and affordable housing modules (PHP, Python, PostgreSQL, JavaScript); built income limits import processing 1.6M HUD/Tax Credit records.
- Led cloud migration of client documents to AWS S3; redesigned document storage database architecture migrating hundreds of thousands of records with 6 months post-deployment support.

PROJECTS

Leasel: AI Document Intelligence & Support Investigation Platform 2026

Three-surface enterprise AI platform for affordable housing compliance: (1) Document Intelligence with OCR, RAG, and 100% citation correctness; (2) 8-agent LangGraph support investigation pipeline with HITL escalation; (3) Tenant Self-Service with hard compliance boundaries. Achieved 93.6% precision on 1,240 lease PDFs across 5 US states.

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · OCR · SQLite · Streamlit

AgentProbe: Agentic AI Testing Toolkit Live on GitHub

Open-source AI agent developer platform: 3-agent LangGraph runtime, CI-integrated 5-metric LLM-as-judge evaluation engine (blocks failing runs pre-production), multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama), ChromaDB RAG pipeline, and React observability dashboard.

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Docker · GitHub Actions

AI on FHIR: NLP-Driven Medical Records Query System 2024

Reduced manual clinical retrieval time ~60% by building a Flask + React NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and embeddings - HIPAA-compliant RBAC across structured medical records.

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings

CrashClarity: Large-Scale Road Accident Analysis (104K Records) 2025

Achieved 85%+ accuracy on 104K+ UK accident records using Random Forest + Logistic Regression with feature engineering across weather, road, and time variables - interactive dashboard deployed on GitHub Pages.

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages

EDUCATION

M.S. Data Analytics Engineering · George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering · Jawaharlal Nehru Technological University — 2017

SKILLS

Languages: Python, PHP, JavaScript, TypeScript, SQL, PL/SQL, R, Bash, HTML/CSS, jQuery, AJAX

AI & Agents: LangChain, LangGraph, AI Agents, Multi-Agent Orchestration, RAG Pipelines, Prompt Engineering, LLM Evaluation, LLM-as-Judge, HITL, ChromaDB, Vector Databases, Embeddings, NLP, Hallucination Detection, OpenAI API, Anthropic API, Groq, Ollama, FastAPI, GenAI, LLMOps, Agent Observability

ML & Data Analytics: PyTorch, TensorFlow, Scikit-learn, XGBoost, Random Forest, Logistic Regression, Isolation Forest, Feature Engineering, Statistical Modeling, Regression Analysis, Anomaly Detection, NumPy, Pandas, spaCy, BERT, Hugging Face, Jupyter, Tableau, Power BI, ggplot2

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), Microservices, PostgreSQL, MySQL, Oracle, RabbitMQ, Redis, Libsodium, Git, Linux, Agile/Scrum

GTM & Integrations: Salesforce, Clay, n8n, Zapier, CRM data pipelines, workflow automation, REST APIs